A Multimodal Stress-Test Corpus for Production-Grade Retrieval-Augmented Generation Pipelines

Benchmarking Layout-Aware Parsers Against Adversarial Document Geometry

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Abstract. We present a deliberately adversarial document designed to expose failure modes in modern document-understanding pipelines, including *Docling*, *Unstructured*, *LlamaParse*, and *Marker*. The corpus combines two-column reflowing text with full-width interruptions, deeply nested tables featuring merged cells and rotated headers, embedded raster figures with multi-line captions, code listings, in-line mathematical notation, multilingual passages (English, Français, Ελληνικά, Русский, 日本語, العربية), floating side-bars, footnotes, and a landscape-oriented appendix. We argue that any pipeline targeting production retrieval-augmented generation (RAG) must correctly recover the **logical reading order**, **semantic structure**, and **tabular relationships** present in such documents — failures in any of these dimensions cascade into retrieval errors and ultimately into hallucinated answers downstream.

Keywords: document AI · layout analysis · table extraction · reading order · OCR · retrieval-augmented generation · benchmark

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1 Introduction

Document understanding has matured rapidly over the past three years, with open-source toolchains such as **Docling** [1], **Unstructured.io** [2], and **Marker** [3] now serving as the ingest layer for the majority of production retrieval-augmented generation (RAG) systems. Despite this progress, real-world documents — invoices, prospectuses, scientific PDFs, regulatory filings — continue to surface long-tail failure modes that benchmarks built on academic papers alone fail to surface.

1.1 Motivation and Scope

This document is, by design, hostile to naïve readers. It interleaves single- and double-column flows, scatters footnotes[1] across page boundaries, embeds raster figures with multi-paragraph captions, and includes tables whose headers span multiple rows and whose cells contain in-line bullet lists. A parser that recovers the logical reading order correctly on this document is — empirically — within striking distance of production readiness.

We use the present document to evaluate three concrete capabilities: **(i)** column-aware text-flow recovery; **(ii)** hierarchical table extraction including merged and rotated cells; and **(iii)** figure–caption association across page breaks.

Reading order the linear sequence in which a human would normally read the visible glyphs on a page. Multi-column layouts produce non-trivial reading orders.

Semantic structure the document's hierarchy of titles, sections, paragraphs, lists, tables, and figures, independent of visual styling.

Tabular relationship the mapping between a cell's value and its header(s), potentially across merged spans.

1.2 Related Work

Prior benchmarks — DocBank [4], PubLayNet [5], and DocLayNet [6] — focus on bounding-box localisation. We complement these with an emphasis on *end-to-end logical recovery*, scoring not the boxes but the downstream answer quality on questions whose answers are split across columns, tables, and figures.

> *“A document parser that cannot recover reading order across a two-column floating side-bar is, for production RAG, no parser at all.”*
>
> *— anonymous reviewer, ICDAR 2024 rebuttal*

[1] Footnotes in this document are intentionally placed at irregular vertical positions to test cross-region association.

2 Two-Column Body with Embedded Artefacts

2.1 Multi-Script Text Rendering

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

English: The quick brown fox jumps over the lazy dog.
Français: Portez ce vieux whisky au juge blond qui fume.
Ελληνικά: Ξεσκεπάζω την ψυχοφθόρα βδελυγμία.
Русский: Съешь же ещё этих мягких французских булок, да выпей чаю.
日本語: いろはにほへとちりぬるをわかよたれそつねならむ。
العربية: نص عربي للاختبار من اليمين إلى اليسار.

2.2 Inline Mathematics and Footnotes

Consider the standard softmax attention[2] over query Q, key K, and value V matrices: $Attention(Q,K,V) = softmax(QK^T/\sqrt{d_k})V$. The denominator $\sqrt{d_k}$ prevents the inner product from growing with dimensionality d_k. For a sequence of length n, the complexity is $O(n^2 \cdot d_k)$, motivating sparse and linear-attention variants such as Performer and Longformer.

Key adversarial features of this section:

- Column reflow with embedded RTL Arabic glyphs
- Mixed sub/superscripts that look like footnote anchors
- A side-bar (see right) that interrupts reading order
- Footnote markers ([2], [3]) whose targets sit on a different page

■ Note — Reading Order Hazard

This side-bar appears *inside* the left column but its content is logically independent. A correct parser should emit it as its own block rather than splicing it into the surrounding paragraph. The side-bar is intentionally narrow and ends with a coloured footer to mimic real-world call-outs.

Severity: HIGH

Continuing after the side-bar, the text resumes its natural flow. Note that a parser that emits the side-bar's content interleaved with this paragraph would corrupt downstream chunks. Furthermore, the transition from the side-bar back to body text may be ambiguous to purely geometric segmentation algorithms.

We continue the discussion to encourage column-spill from the left frame into the right one. The principal benefit of column-aware extraction is that downstream chunkers see semantically coherent passages rather than alternating fragments from two unrelated columns. The same logic applies to footnotes, which must be attached to their host paragraph, not to whichever paragraph happens to lie directly above them in the rendered geometry.

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Figure 3: Infrastructure Composition

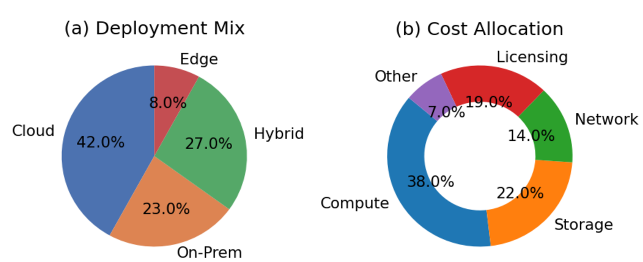

Figure 3. *Infrastructure composition: (a) deployment mix and (b) cost allocation. Note the two-subplot layout, which is challenging*

for figure-detection models that assume a single bounding box per figure.

We continue the discussion to encourage column-spill from the left frame into the right one. The principal benefit of column-aware extraction is that downstream chunkers see semantically coherent passages rather than alternating fragments from two unrelated columns. The same logic applies to footnotes, which must be attached to their host paragraph, not to whichever paragraph happens to lie directly above them in the rendered geometry.

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[2] Vaswani et al., “Attention Is All You Need,” NeurIPS 2017.

[3] This footnote anchor was originally placed two paragraphs above; the anchor ↑ and target ↓ pair span across the column break.

3 Tabular Stress Cases

Tabular structures are responsible for the majority of parser regressions in production. This section deliberately violates several common heuristics: **(a)** headers span multiple rows; **(b)** the body contains a sub-table; **(c)** one row uses cell-level alignment that differs from neighbouring rows; **(d)** negative numbers appear in parentheses; and **(e)** empty cells are visually indistinguishable from zeros.

3.1 Nested Tables with Merged Cells

Segment	Revenue ($M)					Margin (%)					YoY
	Q1	Q2	Q3	Q4	Total	Q1	Q2	Q3	Q4	FY	
Cloud Platform	412.3	438.7	461.2	503.9	1,816.1	32.1	33.4	34.0	35.2	33.7	+18.4%
Edge & IoT	118.4	127.1	131.8	142.6	519.9	21.0	21.7	22.3	23.1	22.1	+11.2%
Professional Services	84.7	79.3	88.1	95.4	347.5	14.8	13.2	15.0	16.1	14.8	+6.7%
Legacy Systems	(12.4)	(9.8)	(14.2)	(7.1)	(43.5)	—	—	—	—	—	−24.8%
Other / Eliminations		2.1		1.8	3.9						n/a
Total Group	**603.0**	**637.4**	**666.9**	**736.6**	**2,643.9**	**28.6**	**29.4**	**29.8**	**31.0**	**29.7**	**+14.1%**

Table 1. *Synthetic financial summary. Note the merged top-row headers (Revenue, Margin), the row-spanning “Segment” and “YoY Growth” columns, parenthesised negatives, em-dash placeholders for non-applicable values, and visually empty cells in the “Other / Eliminations” row that are not the same as zero. A correct parser must preserve all five of these distinctions.*

3.1.1 Cell-Embedded Sub-Tables

The next table embeds a complete sub-table inside one of its cells. Pipelines that flatten tables to plain CSV typically corrupt this structure entirely.

<table>
<tr><th>Component</th><th>Description</th><th>SLA targets</th></tr>
<tr><td>Ingress LB</td><td>Layer-7 reverse proxy with mTLS termination. Routes by Host header and JWT claim.</td><td><table><tr><th>Tier</th><th>QPS</th><th>p99 (ms)</th></tr><tr><td>Bronze</td><td>1k</td><td>82</td></tr><tr><td>Silver</td><td>5k</td><td>47</td></tr><tr><td>Gold</td><td>20k</td><td>19</td></tr></table></td></tr>
<tr><td>Object Store</td><td>S3-compatible, region-replicated, eventual consistency for cross-region reads.</td><td>99.99% availability
11 9s durability
p50 GET: 9 ms</td></tr>
<tr><td>Vector Index</td><td>HNSW with M=32, efConstruction=200. Re-indexed nightly from CDC stream.</td><td>Recall@10: 0.94
Query p99: 35 ms
Index size: 142 GB</td></tr>
</table>

Table 2a. *Component inventory with an embedded sub-table in the “SLA targets” column for “Ingress LB”. The other two cells in that column use formatted text rather than a nested table, mixing representations within a single column.*

doi:10.48550/arXiv.2409.13771

3.2 Rotated Column Headers

Densely packed tables frequently rotate column headers 90° to fit. Geometric OCR followed by naïve token grouping typically reads these as mojibake. The table below combines rotated headers with a left-side row-grouping column.

Group	Engine	Latency	Throughput	Memory	GPU util.	Cost / req	Open-source
Dense	vLLM	★★★★	★★★★★	★★★	★★★★	$0.0021	Yes
	TensorRT-LLM	★★★★★	★★★★	★★★★	★★★★★	$0.0017	Partial
	TGI	★★★	★★★	★★★★	★★★	$0.0029	Yes
MoE	DeepSpeed	★★★★	★★★★	★★	★★★	$0.0034	Yes
	SGLang	★★★★	★★★★★	★★★	★★★★	$0.0022	Yes
Sparse	Triton+FT	★★★	★★★	★★★★★	★★★	$0.0019	Partial

Table 2b. *Specification matrix with rotated column headers and left-column row-group spans (Dense / MoE / Sparse). Unicode star glyphs (★) are used as ordinal indicators; some parsers strip these.*

4 Figures, Charts, and Captions

We embed five PNG figures rendered from matplotlib. Each is preceded by exposition text and followed by a multi-paragraph caption.

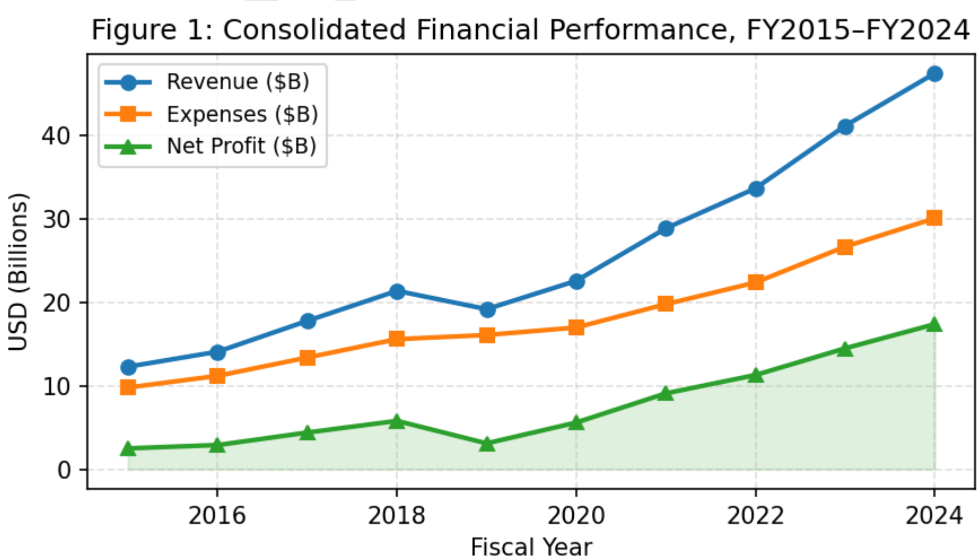

Figure 1. *Consolidated revenue, expense, and net-profit trajectory across FY2015 through FY2024. The shaded region under the profit curve is rendered as a separate graphics primitive and may be detected as a distinct visual element by region-proposal models.*

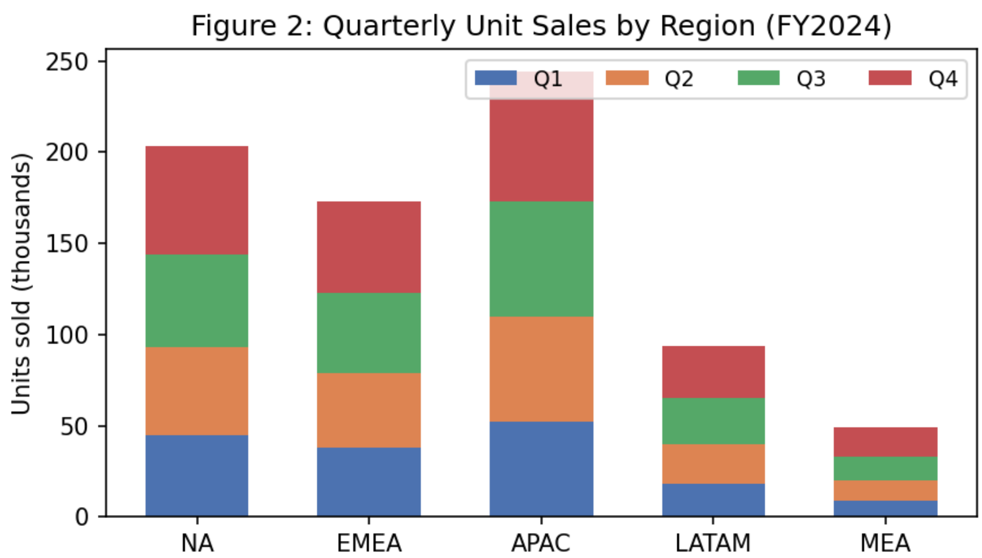

Figure 2. *Stacked-bar visualisation of quarterly unit sales by geographic region. The legend is rendered above the plot area, which breaks the common "legend-inside-axes" assumption used by many chart-extraction heuristics.*

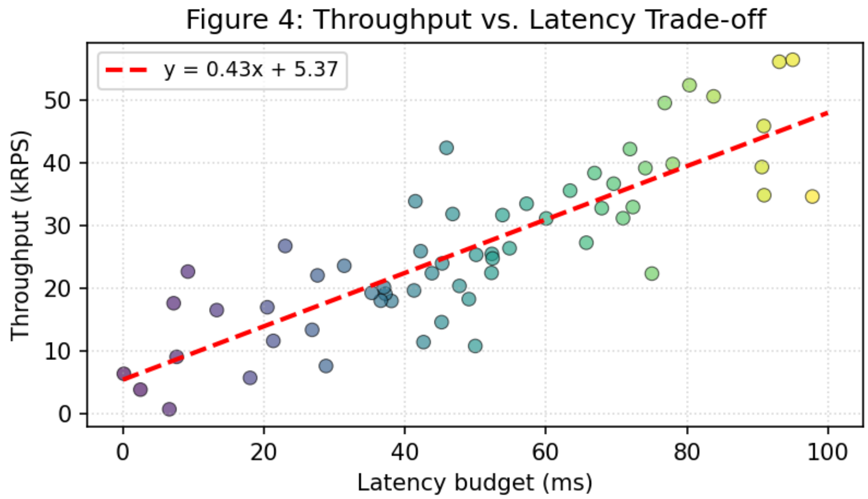

Figure 4. *Scatter plot of throughput against latency budget for the production fleet (n = 60). The dashed regression line and its equation overlay form a composite figure; the equation is rendered as a text glyph, not as a separate annotation layer.*

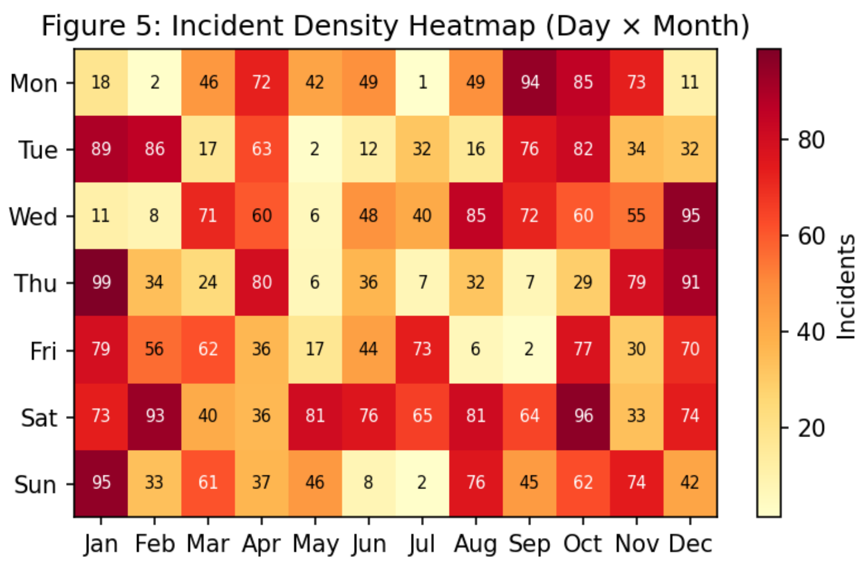

Figure 5. *Day-of-week × month heat-map of operational incidents across FY2024. The colour-bar on the right and the per-cell numeric annotations together constitute three semantic layers (geometry, colour scale, scalar value) that a complete parser should recover.*

5 Code Listings and Pre-formatted Text

Code blocks are typically rendered with a monospaced font, fixed-width padding, and a coloured background. They should be recovered as a single semantic unit, with whitespace preserved exactly.

```
from docling.document_converter import DocumentConverter
from docling.datamodel.base_models   import InputFormat

converter = DocumentConverter(
    allowed_formats=[InputFormat.PDF],
    # Enable table-structure recovery
    pdf_pipeline_options={
        'do_ocr': True,
        'do_table_structure': True,
        'table_structure_options': {
```

```
            'mode': 'accurate',  # vs. 'fast'
            'do_cell_matching': True,
        },
    },
)

result = converter.convert('stress_test_rag.pdf')
for elem in result.document.iterate_items():
    print(elem.label, repr(elem.text)[:80])
```

Listing 1. *Minimal Docling invocation for the present document. Note the indentation hierarchy and the in-line comment.*

```
-- Recover document chunks for retrieval
WITH ranked AS (
    SELECT chunk_id, doc_id, page_no, bbox,
           ts_rank_cd(tsv, query, 32) AS rank,
           1 - (embedding <=> $1::vector) AS cos_sim
    FROM   chunks, plainto_tsquery('english', $2) query
    WHERE  doc_id = ANY($3::uuid[])
)
SELECT * FROM ranked
ORDER  BY 0.4 * rank + 0.6 * cos_sim DESC
LIMIT  20;
```

Listing 2. *Hybrid lexical + vector retrieval query. SQL comments begin with -- and must not be confused with em-dashes in the surrounding prose.*

```
┌──────────┐   ┌──────────┐   ┌──────────┐
│  PDF in  │──▶│  Parser  │──▶│ Chunker  │
└──────────┘   └────┬─────┘   └────┬─────┘
          │          │
          ▼          ▼
     ┌──────────┐   ┌──────────┐
     │  Tables  │   │ Embedder │
     └──────────┘   └────┬─────┘
               ▼
          ┌──────────┐
          │  Vector  │
          │  Store   │
          └──────────┘
```

Listing 3. *ASCII pipeline diagram using box-drawing Unicode characters. Parsers that fall back to Latin-1 will corrupt this block.*

6 Landscape Appendix: Wide Financial Table

This appendix is rotated to landscape orientation to accommodate a wide multi-currency table. Page-rotation handling is a frequent source of bugs in document parsers — the page coordinate system changes, but the underlying text streams may or may not be re-encoded.

Currency	Region	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23	Q1'24	Q2'24	Q3'24	Q4'24	FY22	FY23	FY24
USD	Americas	124.9	126.4	140.2	141.9	149.7	134.4	145.1	138.8	146.3	156.6	147.7	155.4	533	568	606
EUR	EMEA	125.1	139.1	150.8	138.6	165.3	170.2	168.4	171.0	203.4	193.3	194.3	203.6	554	675	795
GBP	EMEA	166.4	172.7	176.0	197.7	189.2	204.3	224.0	227.8	248.3	249.6	267.4	252.1	713	845	1,017
JPY	APAC	67.1	71.2	72.0	76.8	84.1	83.3	86.3	86.9	90.8	104.4	103.3	106.2	287	341	405
CNY	APAC	60.5	66.3	77.0	77.1	80.4	86.7	93.8	98.1	94.9	97.0	107.5	112.7	281	359	412
INR	APAC	73.7	72.0	81.2	83.1	96.2	95.6	98.8	105.2	118.1	120.1	135.0	151.2	310	396	524
BRL	LATAM	103.6	99.0	95.3	97.5	101.6	114.0	120.6	117.3	113.9	123.8	140.4	134.8	395	453	513
MXN	LATAM	162.2	193.6	204.6	212.6	216.3	244.2	238.1	267.1	284.8	327.2	343.8	331.6	773	966	1,287
CAD	Americas	117.3	120.0	112.8	122.6	125.5	119.9	136.1	135.3	144.5	143.0	134.7	146.4	473	517	569
AUD	APAC	45.3	48.0	47.1	48.2	58.6	63.2	58.7	66.9	66.6	79.4	84.8	81.6	189	247	312
TOTAL	**—**	**1,046**	**1,108**	**1,157**	**1,196**	**1,267**	**1,316**	**1,370**	**1,414**	**1,512**	**1,594**	**1,659**	**1,676**	**4,508**	**5,367**	**6,440**

Table 3. *Multi-currency quarterly revenue across FY2022–FY2024 (local-currency millions). Twelve quarterly columns plus three annual summary columns; the FY columns are visually offset by a yellow tint and a darker left-border. The trailing TOTAL row uses a typographic em-dash for the non-aggregable Region cell.*

Appendix A Glossary

BBox. An axis-aligned bounding rectangle (x_0, y_0, x_1, y_1) used to describe the spatial extent of a document element.

Chunk. A retrieval-time unit of text, typically 200–800 tokens, produced by splitting the parsed document along semantic boundaries.

CER. Character Error Rate — Levenshtein distance normalised by the reference length; the standard accuracy metric for OCR.

Docling. An open-source document conversion toolkit developed by IBM Research, supporting layout-aware extraction from PDF, DOCX, and image inputs.

HNSW. Hierarchical Navigable Small World — a graph-based approximate nearest-neighbour algorithm widely used for vector search.

MoE. Mixture-of-Experts — a sparsely-activated neural network architecture in which a routing function selects a small subset of expert sub-networks for each input.

OCR. Optical Character Recognition — the process of converting rasterised glyphs into machine-readable text.

Reading order. The linear sequence in which a document's visible elements should be read; non-trivial for multi-column or floating-element layouts.

RAG. Retrieval-Augmented Generation — a paradigm in which a generative model is conditioned on documents retrieved at inference time.

Span (table). A header cell that occupies multiple rows or columns; encoded in HTML as rowspan / colspan.

TEDS. Tree-Edit-Distance-based Similarity — the standard accuracy metric for table-structure recovery.

Token. A model-specific unit of text, typically a subword fragment produced by a BPE or unigram tokenizer.

Appendix B References

[1] Auer, C., et al. *Docling: An Efficient Document Conversion Toolkit.* arXiv:2408.09869, 2024.

[2] Unstructured.io. *Unstructured: Open-source pre-processing tools for unstructured data.* 2023. https://github.com/Unstructured-IO/unstructured

[3] Paruchuri, V. *Marker: Convert PDF to markdown quickly with high accuracy.* 2024.

[4] Li, M., et al. *DocBank: A Benchmark Dataset for Document Layout Analysis.* COLING 2020.

[5] Zhong, X., Tang, J., Yepes, A. J. *PubLayNet: largest dataset ever for document layout analysis.* ICDAR 2019.

[6] Pfitzmann, B., et al. *DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis.* KDD 2022.

[7] Vaswani, A., et al. *Attention Is All You Need.* NeurIPS 2017.

[8] Smock, B., Pesala, R., Abraham, R. *PubTables-1M: Towards comprehensive table extraction from unstructured documents.* CVPR 2022.

[9] Lewis, P., et al. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* NeurIPS 2020.

[10] Malkov, Y. A., Yashunin, D. A. *Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs.* IEEE TPAMI 2018.